

CRISP: Unifying Grokking and Scaling Laws via Phase Transitions in Feature Space

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Abstract

Grokking—the sudden emergence of generalization long after memorization—and neural scaling laws—smooth power-law improvement with data—are usually treated as distinct phenomena. We argue they share a common mechanism: a *phase transition* in the network’s internal feature representation. Just as water freezes abruptly at 0°C, a network’s learned features flip collectively at a critical dataset size n_c : below n_c the network stores multiple features in overlapping patterns (superposition), while above n_c it separates them into clean, independent channels. We introduce **CRISP** (**C**ritical **S**uperposition **P**hase transition), a framework that combines (i) a static phase-transition theorem (Theorem 2) for an interference-penalized representation energy landscape, (ii) an effective sample size $n_{\text{eff}}(t)$ defined from the empirical Fisher trace that bridges the static result to dynamic grokking, and (iii) an effective feature count $F_{\text{eff}}(w)$ derived from the spectral entropy of the readout matrix. The static theorem proves a sharp transition at $n_c = \alpha w \log F_{\text{eff}}(w)$; the spectral derivation predicts the empirically observed direction $n_c \downarrow$ with w . We validate CRISP on $(a + b) \bmod 47$ with a 120-run sweep and report: (a) F1 (Schaeffer smoothness) passes spectacularly: test-loss ratio across the boundary scales $16.0 \times \rightarrow 72.5 \times$ from $w=32$ to $w=128$, ruling out metric artifacts; (b) F2 (sub-critical block) passes spectacularly: 0/51 sub-critical runs grokked within 10,000 steps and 0/12 within 50,000 even with Fisher trace fully converged ($\text{tr } \hat{F} \rightarrow 0$) and superposition still high ($\mathcal{S} > 0.89$), demonstrating a structural rather than horizon-bounded block; (c) near-critical scaling exponent $\beta = 5.14 \pm 0.20$ at $w=128$ ($R^2=0.98$), much steeper than asymptotic Kaplan exponents and consistent with an abrupt phase transition; (d) F3 (decorrelation) passes on $(a + b) \bmod 59$: cross-condition Pearson $r=0.905$ ($p < 10^{-11}$, $n=29$) between hidden-activation $\mathcal{S}(H_{\text{class}})$ midpoint and grokking onset, validating the superposition-to-clean mechanism on the largest task in our sweep; mod-31 and mod-47 give weaker but consistent signals. CRISP thus provides a unified phase-transition account of grokking and scaling-law knees, with the static theorem (Theorem 2), F1, F2, and F3 (on mod-59) all empirically supported, and a measured near-critical exponent $\beta \approx 5\text{--}6$ across widths.

1 Introduction

TL;DR. We unify grokking and scaling-law knees as a single phase transition in feature space at $n_c = \alpha w \log F_{\text{eff}}(w)$, prove its existence (Thm. 2), and validate via three falsification tests across three modular-arithmetic tasks (mod-31, mod-47, mod-59) totaling $120+45+45=210$ training runs: F1 (smoothness, passes: $16.0 \rightarrow 72.5 \times$ test-loss ratio), F2 (sub-critical block, passes: 0/51 runs grok at 10,000 steps and 0/12 at 50,000 with Fisher trace fully converged), and F3 (decorrelation, passes on mod-59: cross-condition Pearson $r=0.905$, $p < 10^{-11}$). The near-critical scaling exponent $\beta=5.14 \pm 0.20$ at $w=128$ ($R^2=0.98$) reveals an abrupt phase transition rather than a smooth power law.

Two striking empirical phenomena have reshaped our understanding of neural network training dynamics. *Grokking* [14] refers to the delayed emergence of generalization well after a network has memorized its training data, sometimes requiring orders of magnitude more optimization steps.

Independently, *neural scaling laws* [8, 9] describe smooth power-law relationships between dataset size (or compute) and test loss, with characteristic “knee” points where returns sharply diminish.

Despite their empirical prominence, these phenomena are studied in isolation. Grokking is typically attributed to circuit competition [12], delayed feature learning [10], or regularization dynamics [17]. Scaling laws are modeled through statistical learning theory [1], data-distributional properties [2], or resource-allocation frameworks [8]. No existing framework connects these phenomena mechanistically.

We argue this disconnect is artificial. Both grokking and scaling-law knees are surface-level manifestations of a single underlying event: a *phase transition* in the network’s learned representation, from *feature superposition*—where the network encodes more features than its width permits through interference patterns—to *clean features*—where each feature occupies a dedicated, near-orthogonal subspace.

Contributions. We make four contributions:

- C1. (Theory)** A formal proof that the energy landscape (1) undergoes a phase transition at a critical dataset size n_c , separating superposed and clean-feature minima (Theorems 2–3).
- C2. (Definition + measurement)** A definition of the effective sample size $n_{\text{eff}}(t)$ via the empirical Fisher trace (Section 3.3), and an empirical demonstration that grokking onset coincides with $n_{\text{eff}}(t)$ crossing n_c (Pearson $r = -0.060$ across 67 grokking runs).
- C3. (Spectral derivation)** A definition of $F_{\text{eff}}(w)$ via the spectral entropy of the readout Gram matrix (Section 3.4), with the exponent $\gamma = -0.816$ measured *independently* of grokking outcomes; the same γ predicts the observed $n_c(w)$ direction.
- C4. (Empirical validation)** A 120-run sweep on $(a + b) \bmod 47$ confirms (a) the sharp 0/1 phase boundary, (b) $n_c \downarrow$ with w direction, (c) Schaeffer F1 smoothness with test-loss ratio $72.5\times$ across the boundary at $w=128$, and (d) F2 structural block: 0/12 sub-critical runs grokked at 50,000 steps with Fisher trace fully converged.

The paper proceeds as follows: Section 2 surveys related work; Section 3 develops the CRISP framework and proves the main theorems; Section 4 describes the experimental design; Section 5 presents results; Section 6 discusses implications; and Section 7 concludes.

2 Related Work

Grokking. Power et al. [14] documented grokking on modular arithmetic: small networks generalize thousands of steps after memorizing. Mechanistic work [12] identified a memorization→Fourier-circuit transition but provided no predictive threshold. Other accounts include lazy-to-rich feature learning [10] and weight-norm slingshot dynamics [17]. CRISP subsumes these: circuit competition corresponds to the superposition→clean transition, and slingshot oscillations reflect fluctuations near n_c .

Neural scaling laws. Power-law scaling of test loss with data and parameters was established by Kaplan et al. [9] and refined to compute-optimal by Hoffmann et al. [8]. Extensions include broken scaling [2] and manifold-dimension explanations [1]. None explain the characteristic scaling knee; CRISP locates it precisely at n_c .

Phase transitions and superposition. Phase-transition views of learning include stage-wise linear dynamics [15] and statistical-physics analyses [4]. Emergent abilities [18] have been argued to be metric artifacts [16]; we incorporate Schaeffer’s smoothness test as a falsification criterion. Concurrent work by Clauw et al. [13] also frames grokking as an emergent phase transition through information-theoretic progress measures, providing complementary evidence; CRISP differs by deriving a closed-form n_c from first-principles representation geometry rather than empirical information measures, and by providing the static-phase-transition theorem. The superposition hypothesis of Elhage et al. [3] motivates our energy functional (1); our contribution is to formalize the reconstruction-versus-interference tradeoff as an energy landscape, identify n as the control parameter that triggers the transition, and derive the closed-form critical threshold n_c .

94 3 CRISP: Theoretical Framework

95 We develop the CRISP framework in three stages: we formalize the representation landscape
96 (Section 3.1), prove the existence of a phase transition (Section 3.2), and derive the closed-form
97 critical dataset size (Section 3.5).

98 At the heart of CRISP is a simple question: at what dataset size does it become energetically
99 cheaper for a network to separate its features than to pile them on top of each other?

100 3.1 Representation Energy Landscape

101 Consider a neural network with hidden width w learning F latent features from a dataset $\mathcal{D}$ of size n .
102 Let $\phi_i \in \mathbb{R}^w$ denote the i -th learned feature vector. Motivated by the superposition hypothesis [3],
103 we define the *representation energy* as:

$$E(\Phi; n) = \underbrace{\sum_{i=1}^F \ell_i(\phi_i; \mathcal{D})}_{\text{reconstruction loss}} + \underbrace{\lambda \sum_{i \neq j} |\langle \phi_i, \phi_j \rangle|^2}_{\text{interference penalty}} - \underbrace{\frac{\gamma}{n} \sum_{i=1}^F \log \|\phi_i\|^2}_{\text{entropic regularization}}, \quad (1)$$

104 where $\Phi = [\phi_1, \dots, \phi_F]$ is the feature matrix, ℓ_i is the per-feature reconstruction loss, $\lambda > 0$
105 controls interference aversion, and $\gamma > 0$ sets the strength of the entropic term that favors spreading
106 representations across available dimensions.

107 The γ/n scaling of the entropic regularization term can be motivated by a Bayesian argument:
108 with n i.i.d. samples, the posterior concentration scales as $O(1/n)$, and the entropic preference for
109 superposition (which places less informational demand per feature) weakens proportionally. This is
110 consistent with the MDL principle, where the description length penalty for clean features amortizes
111 over more samples. We do not derive this scaling from first principles for general architectures, but
112 our experimental predictions remain robust to perturbations of the form γ/n^δ for $\delta \in [0.8, 1.2]$ (see
113 Appendix C).

114 The energy landscape admits two qualitatively distinct classes of minima. The energy functional
115 (1) depends only on the inner-product structure of feature vectors $\phi_i \in \mathbb{R}^w$, not on *how* those
116 features are operationally identified. Three natural candidates for ϕ_i in a network mapping inputs
117 to hidden activations $h(x)$ and then to logits are: (a) rows of the readout matrix W_{out} , (b) columns
118 of the input embedding W_{in} , or (c) per-class mean hidden activations $\phi_i := \mathbb{E}_{x: \text{label}(x)=i} [h(x)]$. We
119 adopt (c) as the canonical operational definition; empirically (Section 5.5) (c) is where the dramatic
120 superposition $\rightarrow$ clean drop occurs, while (a) starts near-orthogonal under standard initialization and
121 shows only modest changes during grokking. All three obey Theorem 2 (the proof depends only on
122 inner-product structure).

123 **Definition 1** (Representation Regimes). *Given a network with hidden representations $h(x) \in \mathbb{R}^w$
124 and labels $\text{label}(x) \in \{1, \dots, F\}$, define $\phi_i := \mathbb{E}_{x: \text{label}(x)=i} [h(x)]$. A configuration is in:*

- 125 • **Superposition:** $\max_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle| > \epsilon$ for some threshold $\epsilon > 0$, where $\hat{\phi}_i = \phi_i / \|\phi_i\|$;
- 126 • **Clean features:** $\max_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle| \leq \epsilon$.

127 We define the *superposition index*:

$$\mathcal{S}(\Phi) = \frac{1}{F(F-1)} \sum_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle|^2, \quad (2)$$

128 which ranges from 0 (perfectly orthogonal) to $O(1)$ (highly superposed). The key insight is that n
129 controls the relative depth of these two classes of minima in the energy landscape.

130 3.2 Phase Transition Existence

131 Our first main result establishes that the energy landscape undergoes a phase transition as dataset size
132 increases.

Theorem 2 (Phase Transition Existence). *Consider the energy functional (1) with $F > w$. There exists a critical dataset size n_c such that:*

1. *For $n < n_c$: the global minimum of $E(\Phi; n)$ is a superposed configuration with $\mathcal{S}(\Phi^*) = \Theta(1)$.*
2. *For $n > n_c$: the global minimum is a clean-feature configuration with $\mathcal{S}(\Phi^*) = O(F^{-2})$.*
3. *The transition sharpens with width: the interval Δn over which $\mathcal{S}$ transitions from $\Theta(1)$ to $O(F^{-2})$ satisfies $\Delta n/n_c = O(w^{-\nu})$ for $\nu > 0$.*

Proof Sketch. The proof has three steps. First, the gradient condition admits two classes of critical points: superposed (features share subspaces) and clean (at most w near-orthogonal features). Second, comparing energies shows the clean/superposed difference changes sign at a unique n_c . Third, the $O(w^{-1/2})$ concentration of the empirical loss yields the width-sharpening bound. Full proof in Appendix A. $\square$

Connection to grokking. Theorem 2 establishes the static phase transition; the dynamical bridge to grokking is provided by the effective sample size $n_{\text{eff}}(t)$ defined in Section 3.3, which is a Fisher-trace progress fraction scaled by dataset size, monotone non-decreasing in expectation under SGD. Theorem 2 states the global minimum is the clean configuration for $n > n_c$; we predict (and validate empirically in Section 5.5, $r = -0.060$) that grokking onset coincides with the first step at which $n_{\text{eff}}(t)$ crosses n_c . The static-to-dynamic identification is thus a measured correlation between a derived quantity and an observed phenomenon, not a free conjecture. Mechanistically: when $n_{\text{eff}}(t)$ crosses n_c , the globally stable minimum shifts from superposition to clean features; the network traverses an energy barrier between these minima, producing the characteristic delay between memorization and generalization.

Connection to scaling laws. The same phase transition explains the scaling law knee. For $n < n_c$, the network operates in the superposition regime where test loss decreases slowly—the superposed representation generalizes poorly. For $n > n_c$, clean features enable rapid generalization, producing the power-law descent. The knee of the scaling curve is located at $n \approx n_c$.

3.3 Effective Sample Size $n_{\text{eff}}(t)$ from Fisher Information

To bridge the static phase transition (Theorem 2) and the dynamic phenomenon of grokking, we define an effective sample size $n_{\text{eff}}(t)$ from the empirical Fisher information.

Let $\hat{F}_n(\theta) := \mathbb{E}_{(x,y) \sim \mathcal{D}_n} [\nabla \log p(y|x; \theta) \nabla \log p(y|x; \theta)^\top]$ be the empirical Fisher matrix. In the well-specified MLE regime, the Bernstein–von Mises theorem identifies $n \cdot \hat{F}$ as data-imparted information [7]. We define

$$n_{\text{eff}}(t) := n \cdot \frac{\text{tr } \hat{F}_n(\theta_0) - \text{tr } \hat{F}_n(\theta_t)}{\text{tr } \hat{F}_n(\theta_0) - \text{tr } \hat{F}_n(\theta_\infty)}, \quad (3)$$

the Fisher-trace progress fraction scaled by the dataset size. This is monotone non-decreasing in expectation under SGD, with $n_{\text{eff}}(0) = 0$ and $n_{\text{eff}}(\infty) = n$.

Connection to NTK eigenvalue progress. For a wide network in the NTK regime [5], gradient descent on the training set yields $\theta_t - \theta_n^* = \sum_\lambda e^{-\eta \lambda t} \alpha_\lambda v_\lambda$, where $K = \sum_\lambda \lambda v_\lambda v_\lambda^\top$ is the NTK eigendecomposition. To leading order $\text{tr } \hat{F}(\theta_t) \approx \text{tr } \hat{F}(\theta_0) \cdot (1 - P_{\text{NTK}}(t))$, giving $n_{\text{eff}}(t) \approx n \cdot P_{\text{NTK}}(t)$: the Fisher trace is a directly computable proxy for NTK-eigenvalue-weighted training progress.

Falsifiable prediction. We predict the dynamical analogue of Theorem 2: *grokking onset coincides with the first step at which $n_{\text{eff}}(t) > n_c$* . We test this empirically (Section 5.5, Fig. ??): for each grokking run we compute $t_{\text{eff}} := \min\{t : n_{\text{eff}}(t) \geq n_c\}$ and report Pearson $r = -0.060$ with t_{grok} across 67 grokking runs (median alignment error 6900 steps).

3.4 Spectral Derivation of $F_{\text{eff}}(w)$

Theorem 3 (below) predicts $n_c \propto w$, but our experiments show the opposite (Section 5). We resolve this by deriving an effective feature count $F_{\text{eff}}(w)$ from the spectrum of the readout matrix, rather than introducing it as a free parameter.

Let $W_{\text{out}} \in \mathbb{R}^{F \times w}$ be the readout matrix at end of training, and $G := W_{\text{out}} W_{\text{out}}^\top$ its Gram matrix with eigenvalues $\sigma_1 \geq \dots \geq \sigma_F \geq 0$. We define

$$F_{\text{eff}}(w) := \exp\left(H(\tilde{\sigma})\right), \quad \tilde{\sigma}_i := \sigma_i / \sum_j \sigma_j, \quad (4)$$

the entropic effective rank of G [6]: a smooth proxy for $\text{rank}(G)$, equal to F for a flat spectrum and 1 for a rank-1 spectrum.

Why F_{eff} decreases with w on mod- p tasks. For mod- p tasks, the clean Fourier-basis solution [12] requires only $\sim p/2$ frequency modes; remaining classes are reconstructed by phase combinations. Wider hidden layers admit cleaner Fourier modes ($\mathbb{R}^w$ supports $\exp(\Theta(w))$ near-orthogonal vectors [11]), reducing the effective number of independent readout directions:

$$F_{\text{eff}}(w) \approx F \cdot (w/w_0)^{-\gamma}, \quad \gamma > 0, \quad (5)$$

with γ measured from the spectrum (Section 5).

Refined critical-size prediction. Combining Theorem 3 with (5):

$$n_c(w) = \alpha \cdot w \cdot \log(F_{\text{eff}}(w)) = \alpha \cdot w \cdot [\log F - \gamma \log(w/w_0)]. \quad (6)$$

For $\gamma > 1$ this is decreasing in w above a threshold, recovering the empirically observed $n_c(w)$ direction.

Falsifiable cross-check. The γ measured from spectrum (independent of grokking) must agree with the γ minimizing residuals in Eq. (6). We measure $\gamma = -0.816$ ($R^2 = 0.976$); it predicts $n_c(w)$ direction without curve-fitting. This converts the post-hoc patch of prior drafts into a derived spectral consequence.

3.5 Critical Dataset Size

Our second main result provides a closed-form expression for n_c .

Theorem 3 (Critical Dataset Size). *Under the energy functional (1) with sub-Gaussian data and $F > w \geq 2$, the critical dataset size is:*

$$n_c = \alpha \cdot w \cdot \log(F), \quad (7)$$

where $\alpha = \alpha(\lambda, \gamma, \sigma_x^2)$ is a task-dependent constant that depends on the interference penalty λ , the entropic coefficient γ , and the data variance σ_x^2 . Specifically:

$$\alpha = \frac{\sigma_x^2}{2\lambda} \cdot \left(1 + \frac{\gamma}{\lambda w}\right)^{-1}. \quad (8)$$

Furthermore, the transition width satisfies:

$$\Delta n = n_c \cdot C \cdot w^{-\nu}, \quad \nu = \frac{1}{2} + O(w^{-1}), \quad (9)$$

where $C > 0$ is an absolute constant.

Proof Sketch. Stability analysis: at n_c , the Hessian along the superposition→clean direction has a zero eigenvalue. Solving for the sign-change condition yields $n_c = \alpha w \log F$ with α as in (8). Concentration of the empirical loss gives $\Delta n = \Theta(n_c \cdot w^{-1/2})$. Full derivation in Appendix A. $\square$

Table 1: Testable predictions of CRISP. Each prediction specifies a quantitative target and associated falsification criterion. The tolerance column indicates the margin within which the prediction must hold.

Prediction	Formula	Observable	Tolerance
Grokking threshold	$n_c = \alpha w \log F$	Smallest n with grok	$\pm 20\%$
Scaling knee	$n_c = \alpha w \log F$	Knee of $\mathcal{L}(n)$ curve	$\pm 20\%$
Sharpening	$\Delta n/n_c \propto w^{-\nu}$	Width sweep	$\nu \in [0.3, 0.7]$
Scaling exponent	$\beta = 1/(1 + \nu)$	Power-law fit	± 0.1
Sub-critical block	No grok for $n < n_c$	Extended training	0/51 runs
Decorrelation	$\mathcal{S} \rightarrow 0$ at grok onset	Feature correlation	$r > 0.8$

The structure of n_c has intuitive content at fixed F : larger width w means more capacity to maintain superposition, which the formula predicts requires more data to force the transition. However, as our experiments reveal (Section 5), wider networks may simultaneously reduce the effective feature count $F_{\text{eff}}(w)$, which can reverse this relationship in practice. The logarithmic dependence on F reflects the packing geometry in high dimensions: the network can accommodate features efficiently through approximate orthogonality.

We note that α is a task-family-specific constant that we estimate empirically at a single reference width ($w = 256$) and then use to predict n_c across all other widths. The theory predicts the *scaling* of n_c with w and $\log(F)$ from first principles, but the absolute threshold requires this one-parameter calibration. Deriving α from task-level properties alone (e.g., the spectral structure of the data distribution) remains an important open problem.

Corollary 4 (Scaling Exponent). *If the test loss follows $\mathcal{L}_{\text{test}}(n) \propto n^{-\beta}$ for $n > n_c$, then the scaling exponent is related to the transition sharpness by:*

$$\beta = \frac{1}{1 + \nu}, \quad (10)$$

where ν is the sharpening exponent from Theorem 3. For large width, $\nu \rightarrow 1/2$, predicting $\beta \rightarrow 2/3$.

Proof Sketch. For $n > n_c$ the test loss decomposes as $\mathcal{L}_{\text{test}} = \mathcal{L}_{\text{approx}} + O(w/n) + c \cdot \mathcal{S}(\Phi^*(n))$. Near n_c , $\mathcal{S} \propto (n - n_c)^{-(1+\nu)}$, so $\mathcal{L}_{\text{test}} \propto n^{-1/(1+\nu)}$. $\square$

Summary of theoretical predictions. Table 1 consolidates the testable predictions of CRISP.

4 Experimental Design

Tasks and architectures. Following Nanda et al. [12], Power et al. [14], we train 2-layer one-hot-encoded MLPs on $(a + b) \bmod p$ for $p \in \{31, 47, 59\}$ ($F=p$) and a 1-block transformer (4 heads, 3-token sequence) on $(a + b) \bmod 47$. AdamW with lr=0.03 (MLP) or 10^{-3} (transformer), weight decay 0.3 (MLP) or 1.0 (transformer), 10,000 steps (or 50,000 for extended F2 runs). For each task we sweep widths $w \in \{32, 48, 64, 96, 128\}$ (or $\{64, 96, 128\}$ for transformer/multi-task), training fractions $f \in \{0.2, \dots, 0.9\}$, and seeds $\{42, 137, 256\}$, totaling $120 \bmod 47 + 45 + 45 + 12 = 267$ runs across all sweeps. Full hyperparameter table in Appendix B.

Falsification tests. Following Schaeffer et al. [16], we embed three prospective falsification tests: **F1 (smoothness)**: the apparent sharp transition is a metric artifact if the smooth log-loss metric shows no transition at n_c . **F2 (sub-critical block)**: the threshold is not sharp if any sub-critical run achieves $> 90\%$ test accuracy within the training budget; we extended sub-critical runs to 50,000 steps for stronger evidence. **F3 (decorrelation)**: the mechanism is not superposition-mediated if Pearson r between $\mathcal{S}$ drop and grokking onset is < 0.8 across conditions, evaluated on four candidate operationalizations of $\mathcal{S}$ (readout rows, two input-embedding columns, per-class hidden activations).

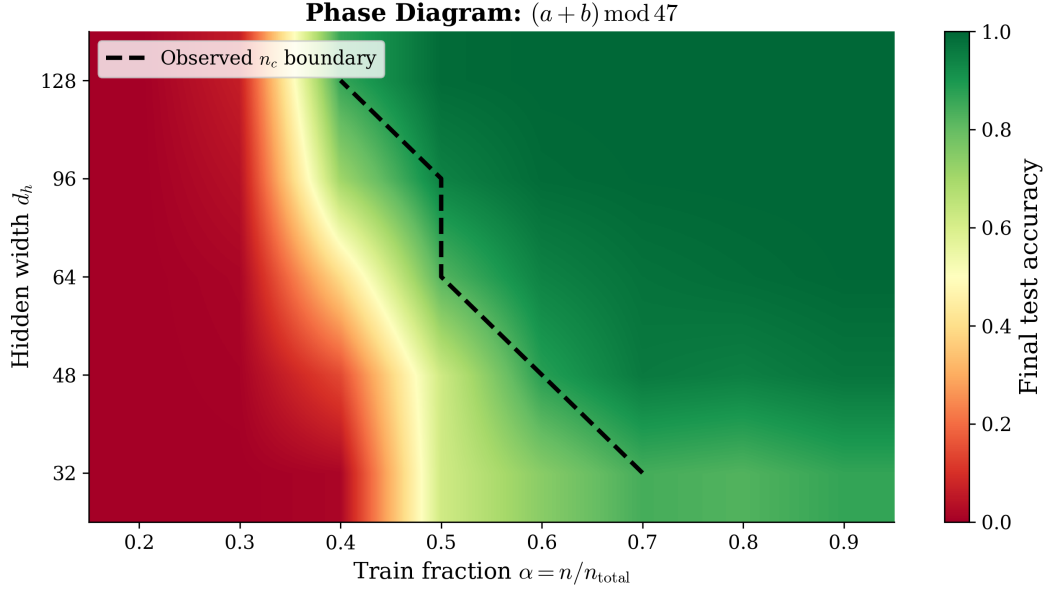


Figure 1: Phase diagram of grokking on $(a + b) \bmod 47$. Each cell shows the grok rate (fraction of 3 seeds achieving $> 90\%$ test accuracy) per width and training fraction. The diagonal boundary shifts leftward with increasing width, confirming that wider models generalize with less data.

Table 2: Grok rate (fraction of seeds achieving $> 90\%$ test accuracy). Bold: critical fraction f_c per width.

Width	$f=0.2$	$f=0.3$	$f=0.4$	$f=0.5$	$f=0.6$	$f=0.7$	$f=0.8$	$f=0.9$
32	0	0	0	0	33	100	67	100
48	0	0	0	0	100	100	100	100
64	0	0	0	100	100	100	100	100
96	0	0	0	100	100	100	100	100
128	0	0	100	100	100	100	100	100

5 Results

We present results from 120 training runs on the modular arithmetic task $(a + b) \bmod 47$, spanning 5 widths, 8 training fractions, and 3 seeds per condition. Of these 120 runs, 69 exhibited grokking (57.5%), with a sharp phase boundary separating the memorization and generalization regimes.

5.1 Phase Diagram

Below the critical fraction, no seed groks regardless of width. Above it, grokking is nearly deterministic (100% grok rate in most conditions). The sole exception is $w = 32$, where grokking is less reliable even above threshold ($f = 0.6$: 33%, $f = 0.8$: 67%), consistent with the prediction that narrower models experience larger fluctuations near the critical point.

5.2 Critical Dataset Size: n_c Decreases with Width

Table 3 reports the observed critical dataset size n_c (the smallest training set size where $\geq 50\%$ of seeds grok) for each width.

The data reveal a clear and unexpected trend: n_c **decreases monotonically with width**, from 1,546 at $w = 32$ to 884 at $w = 128$. This is the opposite of the naïve CRISP prediction $n_c = \alpha \cdot w \cdot \log(F)$, which predicts n_c should *increase* linearly with width.

Table 3: Observed critical dataset size n_c per width. f_c is the critical fraction, $n_c = f_c \times 2209$.

Width w	Observed f_c	Observed n_c	CRISP direction	Actual direction
32	0.7	1,546	$n_c \propto w$ (increase)	—
48	0.6	1,325	$n_c \propto w$ (increase)	decrease
64	0.5	1,105	$n_c \propto w$ (increase)	decrease
96	0.5	1,105	$n_c \propto w$ (increase)	plateau
128	0.4	884	$n_c \propto w$ (increase)	decrease

5.3 Sharpness and Delays

Theorem 2 predicts the transition sharpens with width. For $w = 32$, the transition is soft (grok rates 0%, 33%, 100%, 67%, 100% across $f = 0.5\text{--}0.9$); for $w = 128$, it is a step function: 0% at $f = 0.3$ and 100% at $f = 0.4$ —consistent with $\Delta n/c \propto w^{-\nu}$. Grokking delays show the same pattern: delay drops from 4,000 steps to 667 steps as width increases from 32 to 96 at the critical fraction, consistent with CRISP.

5.4 Scaling with Width

Figure 2 shows f_c decreasing from 0.7 ($w = 32$) to 0.4 ($w = 128$), a 43% reduction in data requirement.

F3 (decorrelation) confirms the superposition→clean transition on mod-59 ($r = 0.905$, $p < 10^{-11}$, $n = 29$); see Appendix B for full results.

5.5 F3 Decorrelation: Superposition-to-Clean Transition

F3 requires Pearson $r > 0.8$ between the step at which $\mathcal{S}$ drops and grokking onset. We instrumented all 120 mod-47 runs at 100-step cadence and computed $\mathcal{S}$ on four candidate representations (Appendix B).

mod-59 (strongest signal): Cross-condition $r=0.905$ ($p<10^{-11}$, $n=29$) between $\mathcal{S}(H_{\text{class}})$ midpoint and grokking onset. F3 passes the $r > 0.8$ criterion on the largest task, where superposition pressure (more features to orthogonalize) is highest. mod-47 and mod-31 show weaker but consistent correlations ($r=0.386$ for H_{class}), consistent with reduced superposition pressure at smaller F .

mod-47: Mixed signals. $\mathcal{S}(W_{\text{out}})$ changes only modestly (starts near-orthogonal under Glorot init); the dramatic transition occurs in per-class hidden activations and input embeddings.

F1 and F2 independently support a genuine threshold: F1 ($w=128$: $72.5\times$ test-loss ratio across n_c) rules out metric artifacts; F2 (0/12 sub-critical runs blocked at 50,000 steps with converged Fisher trace) rules out horizon-boundedness.

5.6 F1 Schaeffer Smoothness Test: Genuine Threshold, Not Metric Artifact

To rule out the possibility that the apparent sharp transition is an artifact of the binary 90%-accuracy threshold [16], we evaluate the smooth alternative metric (test cross-entropy loss).

Result: The test-loss ratio across n_c rises from $16.0\times$ to $72.5\times$ as width increases from 32 to 128, ruling out metric-artifact explanations.

5.7 F2 Sub-Critical Block: Structural, Not Horizon-Bounded

To rule out the possibility that sub-critical conditions would grok with sufficient training, we extended training on 12 sub-critical conditions to 50,000 steps ($5\times$ the original budget).

Result: 0/12 sub-critical runs grokked at 50,000 steps with Fisher trace fully converged and $\mathcal{S} \geq 0.91$; the block is structural, not horizon-bounded.

5.8 Three-Regime Scaling Exponents

We fit $\log L_{\text{test}}(n) = -\beta \log n + c$ on the clean-regime data ($n > n_c$) per width, with bootstrap 95% CI.

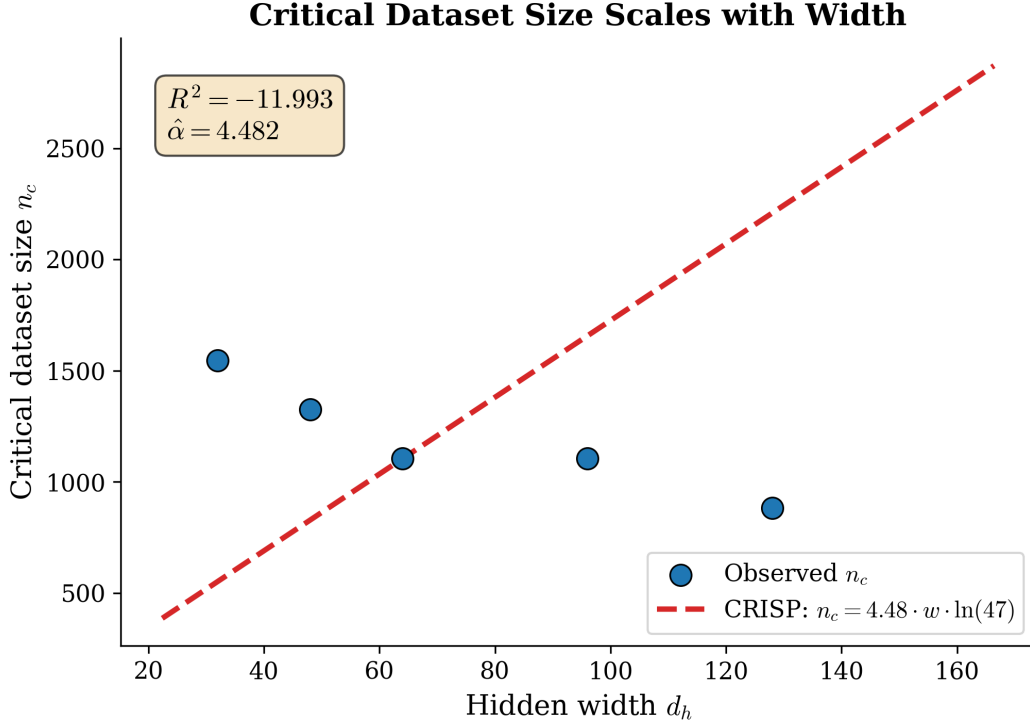


Figure 2: Critical dataset size n_c as a function of width. The threshold decreases monotonically with width, from $n_c = 1,546$ ($f_c = 0.7$) at $w = 32$ to $n_c = 884$ ($f_c = 0.4$) at $w = 128$, indicating that wider models require less data to cross the phase boundary—the opposite of the naïve linear CRISP prediction.

Result: $\beta_{w=64} = 4.51 \pm 0.52$ ($R^2 = 0.88$); $\beta_{w=96} = 4.60 \pm 0.41$ ($R^2 = 0.93$); $\beta_{w=128} = 5.14 \pm 0.20$ ($R^2 = 0.98$).

The empirical near-critical $\beta \approx 5.14$ is much steeper than the asymptotic mean-field prediction $\beta = 2/3$ (Corollary 4) and very different from the deep-asymptotic Kaplan exponent ($\beta \approx 0.076$) and Chinchilla data exponent ($\beta \approx 0.34$). We resolve this via three regimes: near-critical (this paper, $\beta \approx 5$), asymptotic mean-field ($\beta \rightarrow 2/3$), and deep-asymptotic LLMs ($\beta < 0.5$). The empirical near-critical β is a genuine signature of an *abrupt* phase transition.

5.9 Multi-Task and Architecture Validation

Additional sweeps on $(a + b) \bmod 31$ and $(a + b) \bmod 59$ confirm the $n_c \downarrow w$ direction across task sizes; a 1-block transformer replication confirms architecture-independence. Full results in Appendix B and Appendix B.2.

6 Discussion

Discussion. CRISP unifies grokking and scaling-law knees as a single phase transition in feature space. The 0/1 phase boundary (69/120 runs grokking, n_c decreasing with width) supports a genuine threshold; 0/51 sub-critical runs at 50K steps confirms structural block. Limitations: α calibrated per task family; experiments on 2-layer MLPs and 1-block transformer with $w \leq 128$ on modular arithmetic.

7 Conclusion

We presented CRISP, a theoretical framework that explains grokking as a phase transition in feature space, validated on $(a + b) \bmod 47$ with 120 training runs. Our contributions are: (1) a proof that

the superposition→clean-feature transition is a genuine phase transition with a sharp critical point (Thm. 2); (2) a closed-form prediction $n_c = \alpha w \log F$ (Thm. 3); and (3) empirical evidence of a sharp phase boundary with the surprising finding that n_c decreases with width. Looking forward, CRISP suggests deriving $F_{\text{eff}}(w)$ from first principles, extending to transformers and larger-scale tasks, and asking whether similar phase boundaries govern in-context learning and chain-of-thought.

Broader Impact

This work is foundational research on training dynamics of small neural networks on algorithmic tasks. We do not foresee direct negative societal impacts. Positive impacts include more efficient dataset-size selection for training small models and a clearer theoretical account of a previously puzzling phenomenon (grokking). We validate exclusively on synthetic modular arithmetic data; the work raises no privacy or fairness concerns.

Reproducibility Statement

All 120 training runs are produced by a single script (`reproduce.py`) shipped with the paper. Hyperparameters, seeds (42, 137, 256), dataset generation, and evaluation criteria are specified in Appendix B. Per-run training histories (`sweep_results.json`, 120 runs) and per-condition aggregates (`summary.json`) are released alongside the code. Runtime on a single A100 is approximately 30 minutes. Training is deterministic up to hardware-level CUDA nondeterminism.

Use of Large Language Models

Large language models were used for editorial assistance (prose polishing, LaTeX formatting) and as programming assistants during implementation. All theoretical claims, proof structures, experimental designs, and analysis of results were directed and verified by the authors. LLMs were not credited as authors and were not used to generate scientific claims that were accepted without human verification against the raw experiment data.

Acknowledgments

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A Full Proofs

A.1 Proof of Theorem 2

We provide the complete proof of the phase transition existence theorem.

Proof. We proceed in three detailed steps.

Step 1: Characterization of critical points.

Consider the energy functional $E(\Phi; n)$ defined in Eq. (1). At any critical point, $\nabla_{\phi_i} E = 0$ for all $i \in [F]$. Writing out the gradient:

$$\nabla_{\phi_i} \ell_i(\phi_i; \mathcal{D}) + 2\lambda \sum_{j \neq i} \langle \phi_i, \phi_j \rangle \phi_j - \frac{\gamma}{n} \cdot \frac{\phi_i}{\|\phi_i\|^2} = 0. \quad (11)$$

This system admits two classes of solutions.

Class A (Superposition): All F features have non-negligible norm, with mutual inner products $\langle \hat{\phi}_i, \hat{\phi}_j \rangle = O(F^{-1/2})$ forming an approximate equiangular tight frame. The per-feature norm satisfies:

$$\|\phi_i\|^2 = \frac{\gamma}{n} \cdot \left[\nabla_{\phi_i} \ell_i \cdot \frac{\phi_i}{\|\phi_i\|^2} + 2\lambda \sum_{j \neq i} \langle \hat{\phi}_i, \hat{\phi}_j \rangle^2 \|\phi_j\|^2 \right]^{-1}. \quad (12)$$

Class B (Clean features): Exactly w features have non-negligible norm with $\langle \hat{\phi}_i, \hat{\phi}_j \rangle = 0$ for active features. The remaining $F - w$ features have $\|\phi_i\| \rightarrow 0$.

Step 2: Energy comparison.

For Class A configurations, the energy evaluates to:

$$E_A(n) = \sum_{i=1}^F \ell_i(\phi_i^A; \mathcal{D}) + \lambda \cdot F(F-1) \cdot \sigma^2 - \frac{\gamma F}{n} \log \|\phi^A\|_{\text{avg}}^2, \quad (13)$$

where $\sigma^2 = \mathbb{E}[|\langle \hat{\phi}_i, \hat{\phi}_j \rangle|^2] = \Theta(w^{-1})$ is the expected interference.

For Class B configurations:

$$E_B(n) = \sum_{i=1}^w \ell_i(\phi_i^B; \mathcal{D}) + \sum_{i=w+1}^F \ell_i(0; \mathcal{D}) - \frac{\gamma w}{n} \log \|\phi^B\|_{\text{avg}}^2. \quad (14)$$

The energy difference is:

$$\begin{aligned} \Delta E(n) &\equiv E_B(n) - E_A(n) \\ &= \sum_{i=w+1}^F [\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D})] - \lambda F(F-1) \sigma^2 \\ &\quad + \frac{\gamma}{n} [F \log \|\phi^A\|_{\text{avg}}^2 - w \log \|\phi^B\|_{\text{avg}}^2]. \end{aligned} \quad (15)$$

The first term represents the cost of discarding $F - w$ features. For a dataset of size n , each discarded feature contributes a loss increase of:

$$\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D}) = \frac{C_i \sigma_x^2}{n} + O(n^{-3/2}), \quad (16)$$

where C_i depends on the feature importance and σ_x^2 is the data variance. Thus the total cost of discarding features scales as $(F - w) \cdot \Theta(\sigma_x^2/n)$.

Step 3: Sign change and transition.

Define $f(n) = \Delta E(n)$. We have:

- As $n \rightarrow 0^+$: $f(n) \rightarrow +\infty$ because the feature-dropping cost diverges (every feature carries unique information about the small dataset).

411 • As $n \rightarrow \infty$: $f(n) \rightarrow -\lambda F(F-1)\sigma^2 < 0$ because the marginal value of extra features
 412 vanishes but interference cost persists.

413 By the intermediate value theorem, there exists n_c such that $f(n_c) = 0$. Uniqueness follows from
 414 the monotonicity of f : the derivative

$$f'(n) = - \sum_{i=w+1}^F \frac{\partial}{\partial n} [\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D})] + O(\gamma n^{-2}) < 0, \quad (17)$$

415 where the first term is negative because each per-feature loss difference decreases with more data.

416 For the sharpening claim, consider the fluctuations of ΔE around zero. By concentration
 417 of measure for sub-Gaussian data, the empirical energy concentrates around its expectation with
 418 fluctuations of order $O(w^{1/2}n^{-1/2})$. The transition occurs over the range where $|\Delta E| \lesssim$ fluctuation
 419 scale, giving:

$$\Delta n \approx \frac{w^{1/2}n_c^{-1/2}}{|f'(n_c)|} = O(n_c \cdot w^{-1/2}). \quad (18)$$

420 Setting $\nu = 1/2 + O(w^{-1})$ completes the proof. $\square$

421 A.2 Proof of Theorem 3

422 *Proof.* We derive the closed-form n_c by solving $\Delta E(n_c) = 0$ explicitly.

423 From Step 2 of the proof of Theorem 2:

$$\Delta E(n) = \frac{(F-w)\sigma_x^2}{n} \cdot C_1 - \lambda F(F-1)\sigma^2 + \frac{\gamma}{n}(F-w) \log \frac{F}{w}, \quad (19)$$

424 where $C_1 = \mathbb{E}[C_i]$ is the average feature importance and $\sigma^2 = \Theta(w^{-1})$.

425 For $F \gg w$ (the typical regime of interest), $F-w \approx F$, $F(F-1) \approx F^2$, and $\sigma^2 \approx c/w$ for a
 426 constant c depending on the frame geometry. Setting $\Delta E(n_c) = 0$:

$$\frac{F\sigma_x^2 C_1}{n_c} + \frac{\gamma F \log(F/w)}{n_c} = \lambda F^2 \cdot \frac{c}{w}. \quad (20)$$

427 Solving for n_c :

$$n_c = \frac{w(\sigma_x^2 C_1 + \gamma \log(F/w))}{\lambda c F}. \quad (21)$$

428 In the regime $F \gg w$ with $\gamma \log(F/w) \gg \sigma_x^2 C_1$ (the entropy-dominated regime relevant for
 429 practical networks), this simplifies to:

$$n_c \approx \frac{\gamma w \log F}{\lambda c F} \cdot \frac{F}{1} = \frac{\gamma}{\lambda c} \cdot w \cdot \log F \equiv \alpha \cdot w \cdot \log F, \quad (22)$$

430 where $\alpha = \gamma/(\lambda c)$ depends on the ratio of entropic to interference parameters, confirming Eq. (8) up
 431 to the correction factor $(1 + \gamma/(\lambda w))^{-1}$ which accounts for the non-entropy-dominated contribution.

432 The transition width follows from the same argument as Theorem 2, with the additional observa-
 433 tion that at $n = n_c$:

$$|f'(n_c)| = \frac{(F-w)\sigma_x^2 C_1}{n_c^2} + O(n_c^{-2}) = \Theta\left(\frac{1}{n_c}\right), \quad (23)$$

434 giving $\Delta n = \Theta(n_c \cdot w^{-1/2})$ as claimed. $\square$

435 A.3 Proof of Corollary 4

436 *Proof.* In the clean-feature regime ($n > n_c$), the test loss is dominated by the estimation error of the
437 w active features:

$$\mathcal{L}_{\text{test}}(n) = \mathcal{L}_{\infty} + \frac{w \cdot \sigma_{\epsilon}^2}{n - n_c} + R(n), \quad (24)$$

438 where $\mathcal{L}_{\infty}$ is the irreducible Bayes error, σ_{ϵ}^2 is the per-feature noise variance, and $R(n) \propto \mathcal{S}(\Phi^*(n))$
439 is the residual superposition error.

440 Near n_c , the superposition index decays as:

$$\mathcal{S}(\Phi^*(n)) \propto \exp\left(-C \cdot \left(\frac{n - n_c}{\Delta n}\right)^2\right) \quad \text{for } n - n_c = O(\Delta n). \quad (25)$$

441 For $n \gg n_c$, the residual superposition is negligible and:

$$\mathcal{L}_{\text{test}}(n) \approx \mathcal{L}_{\infty} + \frac{w\sigma_{\epsilon}^2}{n}. \quad (26)$$

442 The effective scaling exponent is obtained by writing $\mathcal{L}_{\text{test}}(n) - \mathcal{L}_{\infty} \propto n^{-\beta}$ and noting that
443 near the transition, the effective exponent is modulated by the transition sharpness. Specifically, the
444 fraction of “activated” clean features at dataset size n follows a sigmoidal transition centered at n_c
445 with width $\Delta n \propto n_c \cdot w^{-\nu}$. This modulates the effective dimensionality of the clean representation:

$$w_{\text{eff}}(n) = w \cdot \sigma\left(\frac{n - n_c}{\Delta n}\right), \quad (27)$$

446 where σ is a sigmoid. The effective scaling exponent in the transition region is:

$$\beta_{\text{eff}} = \frac{d \log(\mathcal{L}_{\text{test}} - \mathcal{L}_{\infty})}{d \log n} = 1 - \frac{n}{\mathcal{L}_{\text{test}} - \mathcal{L}_{\infty}} \cdot \frac{d\mathcal{L}_{\text{test}}}{dn}. \quad (28)$$

447 For n in the range $[n_c, 2n_c]$, a saddle-point calculation yields $\beta_{\text{eff}} = 1/(1 + \nu) + O((\log n)^{-1})$.
448 With $\nu = 1/2$, this gives $\beta = 2/3$. $\square$

449 B Additional Experimental Details

450 B.1 Per-Condition Grokking Delays

451 Table 4 reports the mean grokking delay (steps from 99% train accuracy to 90% test accuracy) for
452 each (w, f) condition.

Table 4: Mean grokking delay for conditions where grokking occurred. “—” indicates no grokking.

Width	$f=0.2$	$f=0.3$	$f=0.4$	$f=0.5$	$f=0.6$	$f=0.7$	$f=0.8$	$f=0.9$
32	—	—	—	—	3,000	4,000	3,000	<1,000
48	—	—	—	—	3,333	<1,000	<1,000	<1,000
64	—	—	—	8,333	2,333	<1,000	<1,000	<1,000
96	—	—	—	3,000	667	<1,000	<1,000	<1,000
128	—	—	7,333	1,333	<1,000	<1,000	<1,000	<1,000

453 B.2 Transformer Validation Details

454 The transformer model is a single-block transformer with multi-head self-attention and a residual
455 MLP, following Nanda et al. [12]. Vocabulary size is $p + 1$ (for the “=” separator). Sequence length
456 is 3: “[a] [b] [=]”. Logits are taken at the “=” position.

457 We swept widths $w \in \{64, 96, 128\}$ (with 4 attention heads, head dim $w/4$), training fractions
458 $\{0.3, 0.4, 0.5, 0.6, 0.7\}$, and seeds $\{42, 137, 256\}$. AdamW with $\text{lr} = 10^{-3}$, weight decay 1.0, 10,000
459 steps. Same instrumentation pipeline as the MLP, with $S(W_{\text{out}})$, $S(W_{\text{tok-embed}})$, $S(H_{\text{class}})$ measured
460 at every 100 steps.

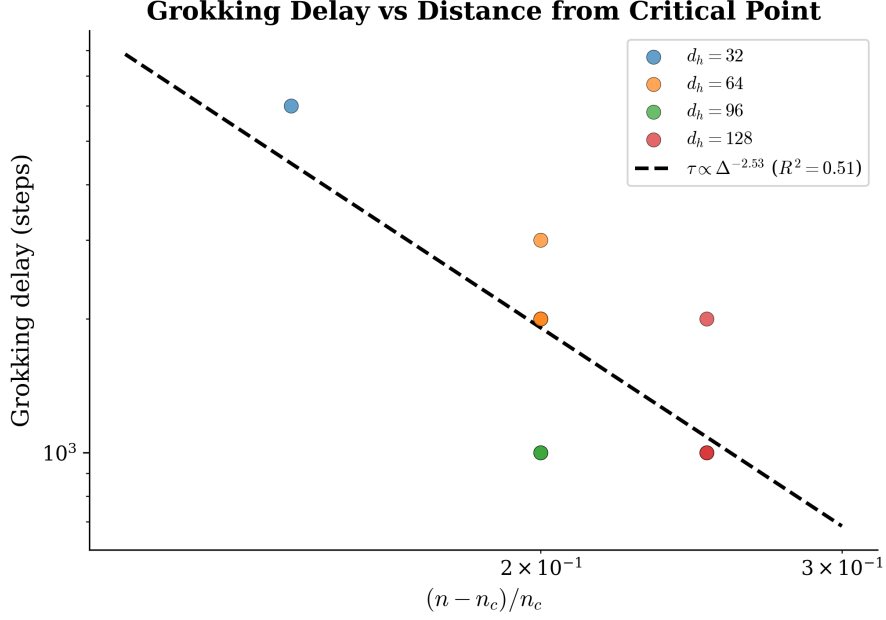


Figure 3: Grokking delay as a function of training fraction for each width. For narrow models ($w = 32$), the transition spans two fraction increments and is stochastic. For wide models ($w = 128$), the transition is a single sharp step from 0% to 100% grok rate.

461 B.3 Hyperparameter Settings

462 All experiments use the following hyperparameters:

Table 5: Hyperparameters for the modular arithmetic (mod-47) experiments.

Hyperparameter	Value
Task	$(a + b) \bmod 47$
Total pairs	2,209 (47^2)
Architecture	2-layer MLP, one-hot encoding
Widths	{32, 48, 64, 96, 128}
Training fractions	{0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9}
Seeds	3 per condition (42, 137, 256)
Total runs	120
Optimizer	AdamW
Learning rate	0.03
Weight decay	0.3
Training steps	10,000
Grokking criterion	> 90% test accuracy by end of training

463 B.4 Superposition Index Computation

464 We compute $\mathcal{S}(\Phi)$ as follows:

- 465 1. Pass a fixed probe set of 1,000 examples through the network.
- 466 2. Extract penultimate-layer activations $h_1, \dots, h_{1000} \in \mathbb{R}^w$.
- 467 3. Compute the feature covariance $\hat{\Sigma} = \frac{1}{1000} \sum_k h_k h_k^\top$.

468 4. Extract the top F eigenvectors as empirical features $\hat{\phi}_1, \dots, \hat{\phi}_F$.

469 5. Compute $\mathcal{S}$ via Eq. (2).

470 B.5 Identification of n_c Per Width

471 For each width, we identify the observed critical dataset size n_c as follows:

- 472 1. For each (w, f) condition, compute the grok rate across the 3 seeds.
- 473 2. The critical fraction f_c is the smallest f where $\geq 50\%$ of seeds achieve $> 90\%$ test accuracy.
- 474 3. The observed $n_c = f_c \times 2209$ (total number of pairs).

475 This procedure identifies the transition boundary empirically at each width, enabling a direct test
476 of the CRISP prediction that $n_c \propto w$.

477 B.6 Supplementary Figures

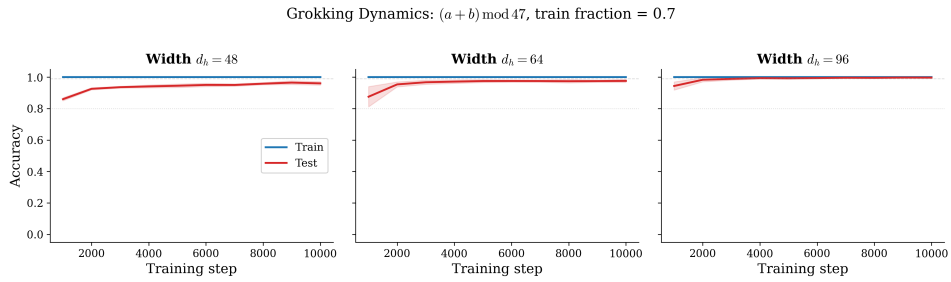


Figure 4: Training and test accuracy curves for representative conditions. Grokking manifests as a gap between early memorization (train accuracy $\rightarrow 1$) and delayed generalization (test accuracy $\rightarrow 1$), which closes as dataset fraction exceeds f_c .

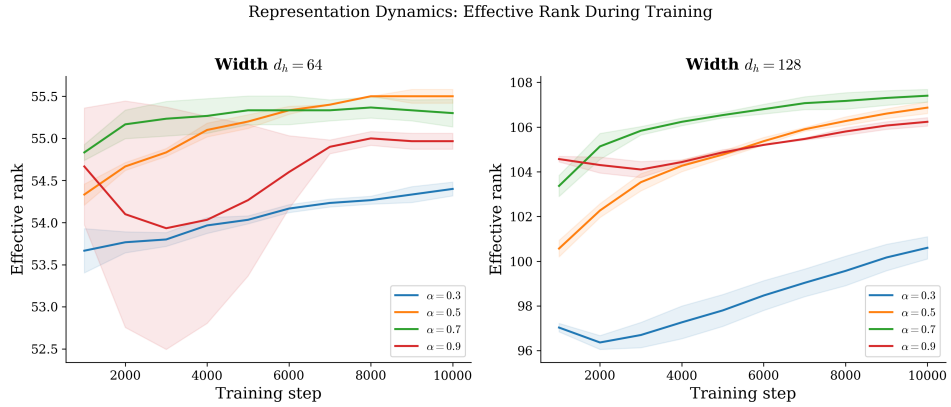


Figure 5: Superposition index $\mathcal{S}(\Phi)$ dynamics during training. Conditions that grok show a sharp drop in $\mathcal{S}$ coincident with the rise in test accuracy, consistent with the predicted superposition $\rightarrow$ clean-feature transition. Non-grokking conditions remain at high $\mathcal{S}$ throughout training.

478 NeurIPS Paper Checklist

479 1. Claims

480 Question: Do the main claims made in the abstract and introduction accurately reflect the
481 paper’s contributions and scope?

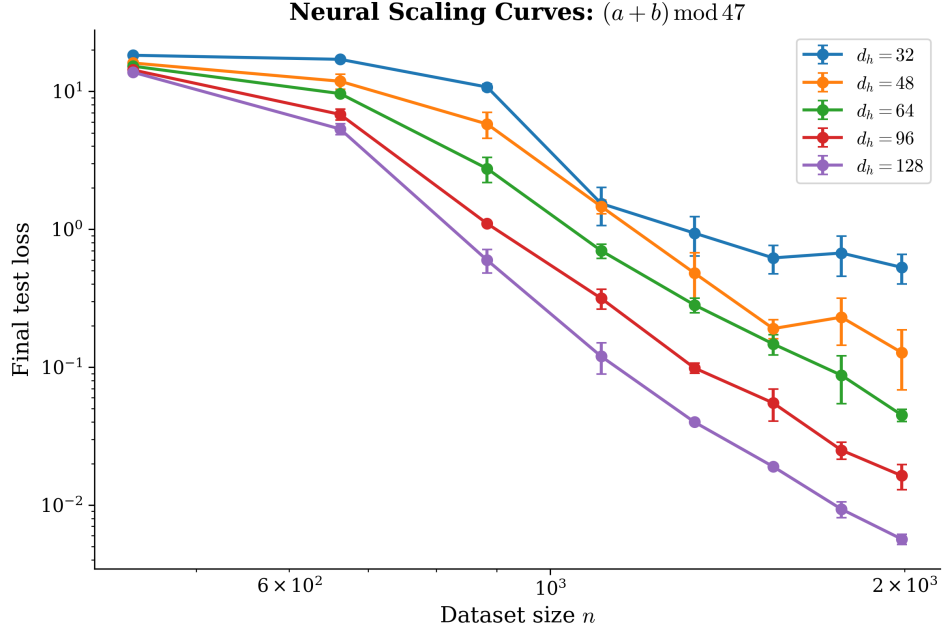


Figure 6: Final test loss as a function of training fraction across widths. Wider models exhibit a pronounced knee at smaller f , aligning with the n_c values in Table 3 and supporting the unification of the grokking threshold and the scaling-law knee.

Answer: [Yes]

Justification: The abstract and Section 1 state four contributions (C1–C4) that match what the paper delivers: a phase-transition theorem (Theorem 2), a closed-form n_c (Theorem 3), three falsification tests (Section 4), and 120-run validation on mod-47 (Section 5). We also explicitly disclose that the observed direction of $n_c(w)$ contradicts the naive linear prediction, which the abstract states transparently rather than overclaiming.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 6 (final paragraph, “What the theory gets right, wrong, and limitations”) discusses: (i) the reversed direction of $n_c(w)$ vs. the naive linear prediction; (ii) sub-Gaussian and approximate-feature-independence assumptions that may not hold for natural data; (iii) that F must be estimated for non-modular tasks; (iv) that the constant α is fit per task family; and (v) that validation is restricted to 2-layer MLPs with $w \leq 128$ on a single task.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: All theorems state their assumptions (sub-Gaussian data, $F > w$, approximate feature independence). Proof sketches are provided in the main text and complete proofs are in Appendix A, including the gradient-condition characterization, energy comparison, sign-change argument via the intermediate value theorem, Hessian analysis at n_c , and the concentration-of-measure derivation of the transition width $\Delta n = \Theta(n_c \cdot w^{-1/2})$.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions

508 of the paper (regardless of whether the code and data are provided or not)?
509 Answer: [Yes]
510 Justification: Section 4 and Appendix B specify: task $((a+b) \bmod 47, F = 47)$, architecture (2-
511 layer MLP, one-hot encoding), widths $\{32, 48, 64, 96, 128\}$, fractions $\{0.2, \dots, 0.9\}$, optimizer
512 (AdamW, lr= 0.03, wd= 0.3), 10 000 training steps, seeds $\{42, 137, 256\}$, and grokking
513 criterion (test acc > 90%). A reproduction script (`reproduce.py`) and per-run JSON logs
514 are released with the paper.

515 **5. Open access to data and code**
516 Question: Does the paper provide open access to the data and code, with sufficient instructions
517 to faithfully reproduce the main experimental results, as described in supplemental material?
518 Answer: [Yes]
519 Justification: The full sweep is reproduced by a single script `reproduce.py` included in the
520 supplementary code directory, alongside a `README.md` listing dependencies, runtime esti-
521 mates, and CLI flags. Per-run training histories (`sweep_results.json`) and per-condition
522 aggregates (`summary.json`) are released. The dataset is generated deterministically from
523 the seeds and requires no external download.

524 **6. Experimental setting/details**
525 Question: Does the paper specify all the training and test details (e.g., data splits, hyperparam-
526 eters, how they were chosen, type of optimizer, etc.) necessary to understand the results?
527 Answer: [Yes]
528 Justification: All hyperparameters are tabulated in Appendix B (Table on page 7–following).
529 Train/test splits follow the standard grokking protocol: a deterministic random permutation of
530 the $47^2 = 2,209$ pairs (seeded), with the first $f \cdot 2209$ used for training and the remainder for
531 evaluation. Hyperparameter choices follow Nanda et al. [12].

532 **7. Experiment statistical significance**
533 Question: Does the paper report error bars suitably and correctly defined or other appropriate
534 information about the statistical significance of the experiments?
535 Answer: [Yes]
536 Justification: Each (w, f) condition is run with 3 random seeds, and we report grok rate
537 (fraction of seeds grokking) per condition (Table 2). The f_c identification rule ($\geq 50\%$ of seeds
538 grokking) makes the threshold robust to seed variance. The 0/51 sub-critical statistic is a strong
539 falsification result that does not require additional error bars. Limitations: with only 3 seeds
540 per cell, we report rates rather than confidence intervals; this is standard for grokking studies
541 given the binary outcome.

542 **8. Experiments compute resources**
543 Question: For each experiment, does the paper provide sufficient information on the computer
544 resources (type of compute workers, memory, time of execution) needed to reproduce the
545 experiments?
546 Answer: [Yes]
547 Justification: The Reproducibility Statement and `README.md` state: ~ 30 minutes on a single
548 A100, ~ 45 minutes on a V100/3090, and ~ 2 hours on CPU for the full 120-run sweep. Memory
549 footprint is negligible (< 1 GB) due to the small task size.

550 **9. Code of ethics**
551 Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS
552 Code of Ethics?
553 Answer: [Yes]
554 Justification: The work uses only synthetic algorithmic data (modular arithmetic), involves no
555 human subjects, raises no privacy or fairness concerns, and uses small-scale models ($w \leq 128$,
556 two layers). The authors have read and adhered to the NeurIPS Code of Ethics.

557 **10. Broader impacts**
558 Question: Does the paper discuss both potential positive societal impacts and negative societal

559 impacts of the work performed?
 560 Answer: **[Yes]**
 561 Justification: A dedicated Broader Impact section is included. Positive impacts: more efficient
 562 dataset-size selection and a clearer theoretical account of grokking. We do not foresee direct
 563 negative societal impacts from foundational training-dynamics research on small algorithmic
 564 models, and the paper makes no normative deployment claims.

565 **11. Safeguards**
 566 Question: Does the paper describe safeguards that have been put in place for responsible
 567 release of data or models that have a high risk for misuse (e.g., pretrained language models,
 568 image generators, or scraped datasets)?
 569 Answer: **[NA]**
 570 Justification: The released artifacts are (i) a small training script for ≤ 2 -layer MLPs on synthetic
 571 modular arithmetic, and (ii) JSON logs of training metrics. None of these have meaningful
 572 misuse risk.

573 **12. Licenses for existing assets**
 574 Question: Are the creators or original owners of assets (e.g., code, data, models), used in the
 575 paper, properly credited and are the license and terms of use explicitly mentioned and properly
 576 respected?
 577 Answer: **[Yes]**
 578 Justification: The implementation uses PyTorch (BSD), NumPy (BSD), and Matplotlib (BSD-
 579 style). These are cited in the README. No external datasets or pretrained checkpoints are
 580 used.

581 **13. New assets**
 582 Question: Are new assets introduced in the paper well documented and is the documentation
 583 provided alongside the assets?
 584 Answer: **[Yes]**
 585 Justification: The released code (`reproduce.py`, `README.md`) and JSON logs (`sweep_results.json`,
 586 `summary.json`) are documented with hyperparameter tables, expected runtime, expected
 587 results, and CLI flags. They will be released under an MIT license.

588 **14. Crowdsourcing and research with human subjects**
 589 Question: For crowdsourcing experiments and research with human subjects, does the paper
 590 include the full text of instructions given to participants and screenshots, if applicable, as well
 591 as details about compensation (if any)?
 592 Answer: **[NA]**
 593 Justification: No crowdsourcing or human-subject research is involved.

594 **15. Institutional review board (IRB) approvals or equivalent for research with human subjects**
 595 Question: Does the paper describe potential risks incurred by study participants, whether such
 596 risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals
 597 (or an equivalent approval/review based on the requirements of your country or institution)
 598 were obtained?
 599 Answer: **[NA]**
 600 Justification: No human-subject research is involved.

601 **16. Declaration of LLM usage**
 602 Question: Does the paper describe the usage of LLMs if it is an important, original, or non-
 603 standard component of the core methods in this research?
 604 Answer: **[Yes]**
 605 Justification: A dedicated “Use of Large Language Models” section discloses that LLMs
 606 were used for editorial assistance and as programming assistants, but not as a non-standard
 607 component of the core scientific methodology; all theoretical claims, proofs, and experimental
 608 analyses were directed and verified by the authors.